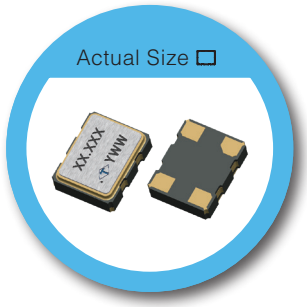
TY-J Type
2.5 X 2.0 mm SMD CMOS Temperature Compensated Crystal Oscillator

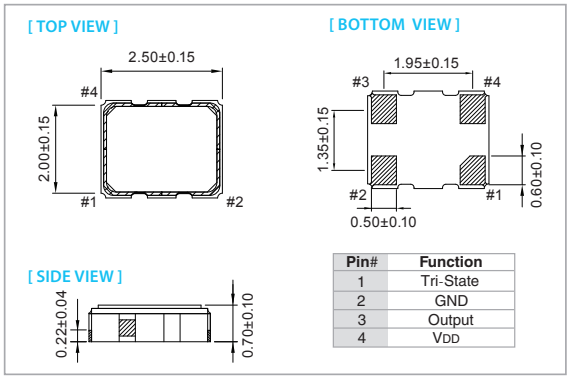


RoHS Compliant

FEATURE
- Conforms to AEC-Q200
- Tolerance: ±2 ppm accuracy @25°C, ±2.5ppm over -40°C to +85°C-
- LVCMOS Output Logic
- Tight symmetry (45 to 55%) available.
- Operation voltage: 1.8V, 2.5V, 3.3V.
- Tri-state enable/disable.
- Femto second phase jitter and -145dBc/Hz at 10kHz offset

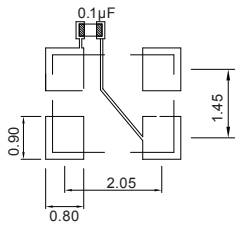
TYPICAL APPLICATION
- Wireless Connectivity
- Smart grid

DIMENSION (mm)



Pin#	Function
1	Tri-State
2	GND
3	Output
4	VDD

SOLDER PAD LAYOUT (mm)



To ensure optimal oscillator performance, place a by-pass capacitor of 0.1µF as close to the part as possible between Vdd and GND pads.

ELECTRICAL SPECIFICATION

Parameter	3.3 V		2.5 V		1.8 V		Unit
	Min.	Max.	Min.	Max.	Min.	Max.	
Supply Voltage Variation (VDD) ±5%	3.135	3.465	2.375	2.625	1.71	1.89	V
Frequency Range	9.5	60	9.5	60	9.5	60	MHz
Supply Current 9.5≦ Fo≦60 MHz	–	10	–	7	–	5	mA
Duty Cycle	45	55	45	55	45	55	%
Output Level (CMOS) Output Hight(Logic"1")	2.97	–	2.25	–	1.62	–	V
Output Low(Logic"0")	–	0.33	–	0.25	–	0.18	
Transition Time : Rise/Fall Time+	–	8	–	8	–	8	nSec
Start Time	–	5	–	5	–	5	mSec
Tri-State(Input to Pin 1) Enable(High voltage or VDD)	2.31	–	1.75	–	1.26	–	V
Disable(Low voltage or GND)	–	0.99	–	0.75	–	0.54	
RMS Phase Jitter (integrated 12kHz ~ 20MHz)	–	1	–	1	–	1	pSec
Phase Noise @ 26MHz 10Hz	–	-80	–	-80	–	-80	dBc/Hz
100Hz	–	-110	–	-110	–	-110	
1kHz	–	-130	–	-130	–	-130	
10kHz	–	-145	–	-145	–	-145	
Aging (@25°C 1st year)	–	±1	–	±1	–	±1	ppm
Storage Temp. Range	-55	125	-55	125	-55	125	°C

Standard frequencies are frequencies which the crystal has been designed and does not imply a stock position
+Transition times are measured between 10% and 90% of VDD, with an output load of 15pF

FREQ.STABILITY vs. TEMP. RANGE

Temp. (°C) \ ppm	±2.5	±5.0	±10.0
-40 ~ +85	○	○	○
-40 ~ +90	△	○	○
-40 ~ +105	×	△	○

* ○: Available △:Conditional X: Not available

Note: not all combination of options are available. Other specifications may be available upon request.

Specifications subject to change without notice.
www.taitien.com
sales@taitien.com.tw